C. Removing Smallest Multiples

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: greedy, math | https://codeforces.com/problemset/problem/1734/C

You are given a set S, which contains the first n positive integers: 1, 2, …, n.

You can perform the following operation on S any number of times (possibly zero):

- Choose a positive integer k where $1 \le k \le n$, such that there exists a multiple of k in S. Then, delete the **smallest** multiple of k from S. This operation requires a cost of k.

You are given a set T, which is a subset of S. Find the minimum possible total cost of operations such that S would be transformed into T. We can show that such a transformation is always possible.

Input

The first line of the input contains a single integer t ($1 \le t \le 10\,000$) — the number of test cases. The description of the test cases follows.

The first line contains a single positive integer n ($1 \le n \le 10^6$).

The second line of each test case contains a binary string of length n, describing the set T. The i-th character of the string is '1' if and only if i is an element of T, and '0' otherwise.

It is guaranteed that the sum of n over all test cases does not exceed 10^6.

Output

For each test case, output one non-negative integer — the minimum possible total cost of operations such that S would be transformed into T.

Example

Input
6 6 111111 7 1101001 4 0000 4 0010 8 10010101 15 110011100101100

Output

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0
11
4
4
17
60
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Note

In the first test case, we shall not perform any operations as S is already equal to T, which is the set {1, 2, 3, 4, 5, 6}.

In the second test case, initially, S = {1, 2, 3, 4, 5, 6, 7}, and T = {1, 2, 4, 7}. We shall perform the following operations:

1. Choose k=3, then delete 3 from S.
2. Choose k=3, then delete 6 from S.
3. Choose k=5, then delete 5 from S.

The total cost is 3+3+5 = 11. It can be shown that this is the smallest cost possible.

In the third test case, initially, S = {1, 2, 3, 4} and T = {} (empty set). We shall perform 4 operations of k=1 to delete 1, 2, 3, and 4.

In the fourth test case, initially, S = {1, 2, 3, 4} and T = {3}. We shall perform two operations with k=1 to delete 1 and 2, then perform one operation with k=2 to delete 4.